

Landscape characteristics explain large-scale variation in demographic traits in forest grouse

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Abstract The effects of landscape composition on species and populations have become increasingly important due to large and rapid habitat changes worldwide. In particular, concern is raised for several forest-dwelling species such as capercaillie and black grouse, because their habitats are continuously changing and deteriorating from human development. Conservation of these species is linked to sustainable forest management that seeks to benefit multiple species, which demands knowledge about demographic rates in relation to forest composition and structure. We related the spatial variation in adult density and chick production of capercaillie and black grouse to landscape characteristics from 13 areas

within the boreal forest of Norway. Linear mixed effects models showed that black grouse and capercaillie had similar associations to landscape characteristics. Adult density of both species was positively related to the proportion of old forest (>80 years), but only if the area had large proportions of mid to high productive forests. Chick production was negatively related to the proportion of old forest, but positively to habitat diversity and more so for black grouse compared to capercaillie. However, the result for chick production suggest that other forest types also are important, and that forest grouse needs a variety of habitats during their life history stages. Management that seeks to simultaneously conserve populations of black grouse and capercaillie needs to ensure a matrix of various forest types. A special focus must be on the critical life history of local populations to successfully preserve viable populations, for black grouse and

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capercaillie this implies protection of old and mid to high productive forest while keeping a heterogeneous landscape.

Keywords Adult density · Chick production · Forest composition · Landscape heterogeneity · *Tetrao tetrix* · *Tetrao urogallus* · Wildlife ecology

Introduction

A central task in nature management and conservation of endangered species is to encourage land use to support biodiversity and viable populations at multiple spatial scales (Pellikka et al. 2005). Temporal and spatial scales have been recognized as important components to fully understand wildlife—habitat relationships (George and Zack 2001). In particular, it is important that conservation of species is based on knowledge from areas large enough to contain habitat types required during all life history stages. This can only be achieved if several life history stages are analyzed to obtain knowledge about how the species in focus utilize habitats during important stages such as breeding or over-wintering. Moreover, environmental stochasticity such as annual fluctuations in weather conditions can affect the relative importance of different habitat types, and thereby underlining the importance of having data over several years to get more reliable species-habitat-relationships. Such data may be hard to obtain for small and threatened populations, and therefore knowledge about the species from areas outside the region where it is endangered will be important and valuable.

Knowledge about species-habitat relationships is especially important in areas with increasing levels of anthropogenic influence, which is recognized as a major threat to global biodiversity (Secretariat of the Convention on Biological Diversity 2010). Forest-dwelling animal species have to cope with rapid changes in habitat structures and permanent habitat loss e.g. due to logging and human development. As a consequence, several forest-dwelling species that previously were abundant over larger areas have become regionally or globally threatened (e.g., Paillet et al. 2010; Kålås et al. 2010). Sustainable forest management that ensures persistence of suitable habitats is essential to avoid extinction, but it is important to keep in mind that several

threatened species may be influenced by conservation actions, and not necessarily in a similar manner. This calls for a multi-species approach when assessing habitat effects on species with particular conservation concern.

The forest-dwelling grouse species capercaillie (*Tetrao urogallus*) and black grouse (*Tetrao tetrix*) are of least concern at a global level (IUCN 2010). However, both species are locally vulnerable and threatened in western and central parts of Europe (Storch 2007), where population declines to a large extent are explained by indirect and direct human impacts. Important factors are habitat degradation due to forestry (Klaus 1991; Kurki et al. 2000; Ludwig et al. 2009), habitat fragmentation leading to increased generalist predators (Storch et al. 2005; Wegge and Rolstad 2011), wildlife fences (Baines and Andrew 2003), and unsustainable harvesting due to hunting during breeding season and poaching (Storch 2007).

Capercaillie and black grouse have a wide distribution in the forested areas in Fennoscandia, and are managed as game species, although a decline in population sizes has been recorded the last two decades (Gregersen and Gregersen 2009). The effects of forestry on grouse ecology have been targeted in many earlier studies, suggesting negative effects of fragmentation (e.g. Kurki et al. 2000). Individual's vital rates are linked to habitat suitability, and most of the previous studies have investigated grouse habitat utilization at local scales (Gjerde and Wegge 1989; Rolstad and Wegge 1989; Helle et al. 1990; Wegge et al. 2005; Wegge and Rolstad 2011). The concern that large areas are needed for populations of these species to be viable has resulted in recent research on how individuals and populations relate to habitat and landscape characteristics at larger spatial scales (Angelstam 2004; Graf et al. 2005; Braunisch and Suchant 2007; Miettinen et al. 2008; Sirkiä et al. 2010). However, in order to better understand how landscape properties affect populations of capercaillie and black grouse, comparisons of areas with contrasting landscape characteristics are needed (Graf et al. 2006, 2007).

Both capercaillie and black grouse depend on landscapes consisting of forest at various succession stages; mainly conifer forests with stands of deciduous trees, as well as bogs (Wegge and Rolstad 2011). Their distribution overlaps to a large extent at the landscape scale, but is to some extent different within the boreal forest according to the successional stage of the forest

stands (Seiskari 1962; Swenson and Angelstam 1993; Wegge and Rolstad 2011). Capercaillie depends on continuous forest habitat with an average seasonal home range of 5 km² (ranging from 1 to 12 km²) (Rolstad et al. 1988; Storch 1995), whereas the habitat used by several individuals in connection with leks corresponded to areas of 20–50 km² (Storch 1995; Rolstad et al. 2009). The home range size of black grouse varies with sex and season from 1 to 20 km² (Caizergues et al. 2003).

Whereas considerable effort have been made in describing the habitat association for black grouse and capercaillie at the local scale, less is known about what determines regional differences in abundance and populations demographic parameters. In this study, we investigated how landscape characteristics influenced adult density and chick production in 13 populations of capercaillie and black grouse in southern and central Norway. The range of landscape types among the populations, the two-species approach, and the inclusion of adult density and chick production allowed us to obtain a more detailed understanding of how habitat composition influences forest grouse populations. Due to the close habitat-relationships in both species (Rolstad and Wegge 1987; Miettinen et al. 2005; Gregersen and Gregersen 2008), we expected that landscape characteristics would be good predictors of demographic parameters. Specifically, we hypothesized that adult density was positively related to the proportion of old forest, and more so for capercaillie compared to black grouse due to that black grouse often is associated more with open habitat (Seiskari 1962). Further we hypothesized that chick production of both species was positively related to the proportion of mid to high productive forest, as this is preferred habitat during breeding season (e.g. Wegge and Rolstad 2011). The heterogeneity of a landscape, characterized by the number of and evenness of proportions of habitat types in a given area, is also characterized by stand size, and we hypothesized that landscape heterogeneity was a better predictor of demographic traits for black grouse compared to capercaillie, as black grouse is found to be more diverse in habitat use (Swenson and Angelstam 1993). Finally, because adult density is more related to autumn and winter survival of sub-adults and adults, we hypothesized that habitat used during these seasons (i.e. old-growth forest) was more important for the estimates of adult density than for chick production.

Methods

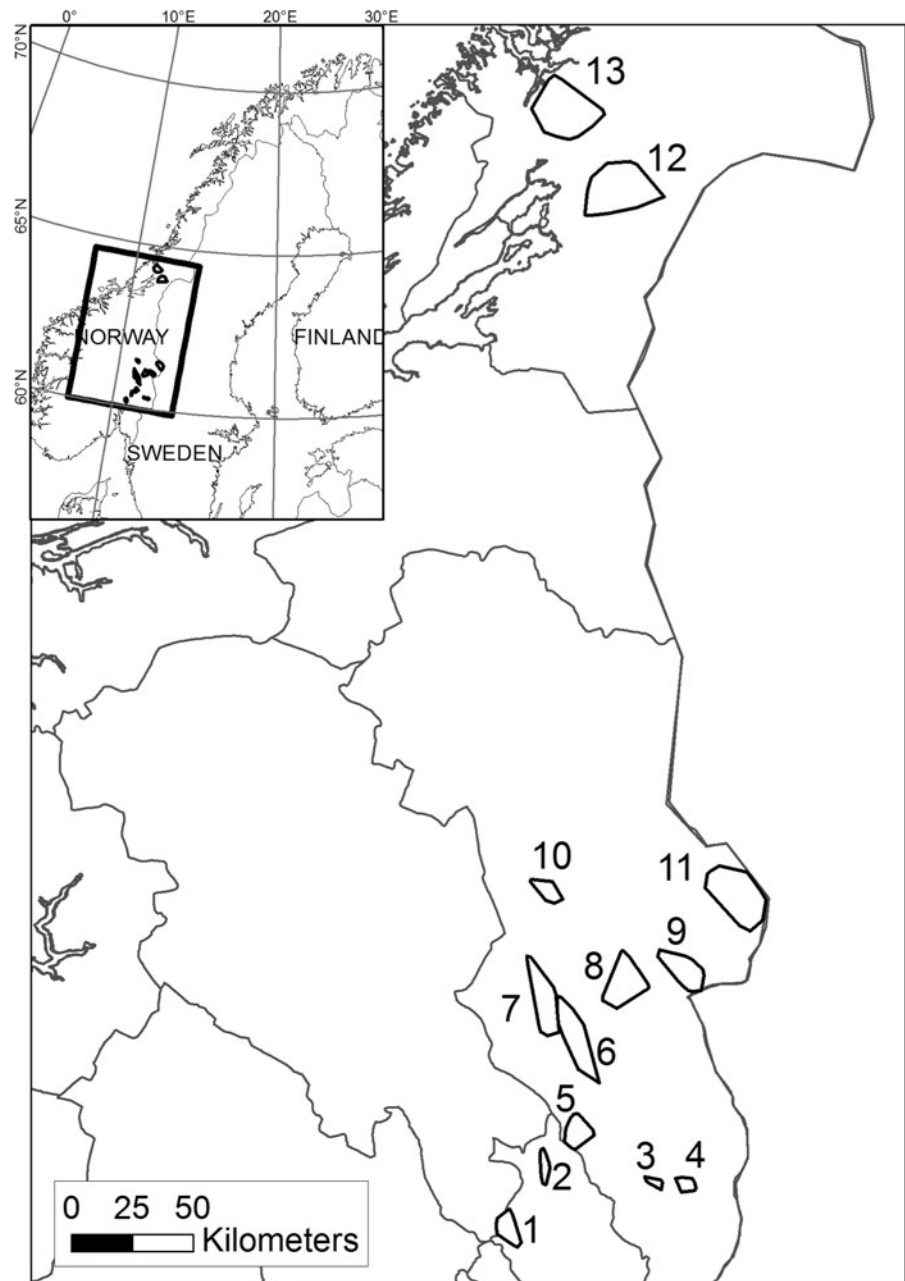
Study areas

The thirteen study areas (range in size: 11–370 km²) were located in Akershus and Hedmark County (area 1–11) and in Nord-Trøndelag County (area 12 and 13; Fig. 1; Table 1). Seven areas were within the continental forests, recognized by large continuous forest areas dominated by Norway spruce (*Picea abies*) and Scots pine (*Pinus sylvestris*) (Moen 1999) with hilly topography impacted by forestry (Puschmann 2005). Five areas were in the low Scots pine forest, typically with low productive conifer forest in combination with mountain birch (*Betula pubescens*) (Moen 1999) and a landscape consisting of varying topography and less intense forestry (Puschmann 2005). One area was within the lower Scots pine heathland region, characterized by hilly, tundra like areas up to 1,500 m above sea level where vegetation had more alpine characteristics (dwarf birch (*B. nana*), heaths and meadows, Puschmann 2005). Human impact in the Scots pine region has traditionally been hunting, fishing and activities linked to free ranging domestic animals, in addition to forestry and agriculture (Puschmann 2005).

Grouse data

Adult density estimates and chick production (chicks per adult female) of capercaillie and black grouse were based on line transect sampling (Buckland et al. 2001) during 2002–2009 in the 13 hunting areas (Fig. 1). Four areas were surveyed in 2002 and by 2007 all 13 areas were surveyed (Table 1). The survey provided data on adult density and chick production for 3 years or more within all areas (Appendix 1 in Supplementary Material), except for one area where estimates of capercaillie and black grouse chick production were only available for 2 years. On average 241 transect lines were surveyed (range 155–302) per year. The transect lines were systematically laid out to represent all habitat types within each hunting area and were on average 3.4 km (range 2–5.6 km). The same sets of lines were surveyed during the whole study period. Each transect line in all areas was surveyed with a minimum of two volunteers and a pointing dog in August each year. Volunteers were recruited and trained to survey (for more details see Solvang et al.

Fig. 1 The 13 forest areas in Norway with capercaillie and black grouse line transect counts used in this study



2009). All estimates were calculated by program Distance (Buckland et al. 2001).

Habitat data

We sought to identify habitat variables related to adult density and chick production, and variables pertaining to habitat use of grouse during important periods of their life history. The habitat variables were mean

forest stand size, proportion of old forest (older than 80 years), proportion of mid to high productive forest (forest producing more than $3 \text{ m}^3 \text{ ha}^{-1} \text{ year}^{-1}$), an index of landscape heterogeneity (SiD; Simpson 1949). See Table 2 for an overview of the habitat variables and their predicted effects of adult density and chick production of forest grouse.

Mean forest stand size, proportion of old forest and proportion of mid to high productive forest were

Table 1 The study areas vary in both size and number of estates (a low number of estates indicates one hunting and management scheme while several may indicate various schemes)

Study area	Size (km ²)	Number of estates	Study period	Mean density (SD) CAP	Mean density (SD) BG	Dominating landscape characteristics (> 60 % cover)
1	78	12	2007–2009	3.0 (1.0)	9.0 (6.9)	N spruce
2	32	25	2007–2009	5.3 (1.5)	6.7 (2.9)	N spruce <80 years
3	11	8	2006–2008	5.0 (2.6)	4.3 (2.1)	N spruce <80 years
4	31	36	2005–2009	5.8 (2.8)	4.4 (2.9)	N spruce <80 years
5	108	9	2002–2009	6.0 (3.4)	7.4 (2.3)	N spruce
6	144	9	2002–2009	4.0 (1.4)	5.8 (2.0)	N spruce <80 years and bogs
7	181	1	2002–2009	3.4 (2.9)	3.6 (2.2)	N spruce and bogs
8	112	141	2002–2009	4.7 (2.3)	10.2 (4.6)	N spruce <80 years and bogs
9	173	47	2005–2009	7.0 (4.3)	9.4 (9.6)	N spruce <80 years and bogs
10	75	6	2005–2008	10.5 (3.3)	9.5 (3.3)	M con
11	272	28	2002–2009	3.8 (3.2)	3.1 (1.5)	M con
12	278	2	2002–2009	2.3 (2.0)	2.6 (2.0)	N spruce and bogs
13	370	3	2003–2009	2.0 (2.0)	2.8 (1.2)	N spruce and bogs

All study areas were studied within 2002–2009, the average density of capercaillie (CAP) and black grouse (BG) with standard deviation (SD) are given and the dominating landscape characteristic(s) covering at least 60 % of the study area. Norway spruce (N spruce) and mixed conifer (m con)

Table 2 The variables for description of the study areas and assessment of adult density and chick production of capercaillie (CAP) and black grouse (BG)

Variable name	Description	Hypothesized effect	Data source
Mean forest stand size	Mean size of forest stands within each study area	Stand size is linked to the fragmentation of the landscape, and BG is assumed to tolerate that better than CAP	SatSkog (Gjertsen 2007)
Mid to high productive forest	Proportion of forest cover producing more than 3 m ³ ha ⁻¹ year ⁻¹	Mid to high productive forests provide more and better forage for chicks of both CAP and BG, increasing the chick survival and recruitment into adult individuals	SatSkog (Gjertsen 2007)
Old forest	Proportion of forest with tree age >80 years	Old forest have habitats more suitable for CAP than BG (e.g. lek sites), particularly important for adult survival	SatSkog (Gjertsen 2007)
SiD (Simpsons index of diversity)	The relative abundance of habitat types (Norway spruce > 80 years, Scots pine > 80 years, Scots pine < 80 years, mixed conifer forest, deciduous forest and bog, Eq. 1). Values close to one means an even distribution of all habitat types, values close to zero suggests a few dominating habitat types.	A heterogeneous landscape provides a variety of habitat types where none are dominating the landscape. This may favor BG more than CAP	SatSkog/AR 50 (National digital maps)

The hypothesized effect is set based on the assumptions described in the “Introduction” section

calculated from SatSkog (Gjertsen 2007). SatSkog were derived from Landsat images (images from 2004 and 2007 with a 15/30 m resolution) that were

classified and verified through the database of national forest inventory and national digital maps (AR50, Aune-Lundberg and Strand 2010). Landscape

heterogeneity was described by the Simpson's index of diversity (SiD), and calculated as:

$$SiD = 1 - \sum_{i=1}^S p_i^2, \quad (1)$$

where p_i is the proportion cover habitat type i and S is the total number of habitat types. The index was based on the following habitat types: Norway spruce younger than 80 years, Norway spruce older than 80 years, Scots pine younger than 80 years, Scots pine older than 80 years, conifer mixed forest, deciduous forest, and bog (see Table 2). Areas with values close to 1 are characterized by an equal distribution of all habitat types, whereas areas with SiD close to 0 have a few dominating habitat types.

Statistical analyses

To evaluate the effect of the habitat variables at landscape scale on adult density and chick production we used linear mixed effects models (Bates et al. 2011a). We included year and study area as random factors in all models. Since grouse dynamics are synchronized over large spatial scales (e.g. Ranta et al. 1995; Kvasnes et al. 2010), area differences in adult density and chick survival caused by different sampling period would be accounted for by the random factor of year. If an area was sampled only in a period where adult densities were low, the random factor of year would account for the effect of sampling period, whereas any other variation was captured by the landscape characteristics. Study area was included as a random factor to account for interdependencies between observations in different years within study areas. By extracting the residual variation attributed to the two random factors, we could also explore how much spatial variation in demographic rates (i.e. between study areas) that were left after accounting for our explanatory (fixed) factors, and accounting for temporal variation by year as a random factor (see below). The global model included all factors and variables.

Adult density and chick production were analyzed separately, but with the same habitat variables and specification of the global model with respect on fixed and random factors. We choose to analyze capercaillie and black grouse adult density and chick production simultaneously to search for species specific effects of

habitat variables. This was done by including interaction terms between the forest characteristic variables and species. In order to evaluate potential combined effects of the forest characteristics, we also included interactions between some of the forest variables, based on the outlined hypotheses (see "Introduction" section). This included proportion of mid to high productive forest in interaction with proportion of old forest, and the SiD in interaction with mean forest stand size. When a model included an interaction, the main effects of the variables in the interaction were always retained in the model. The global model for both adult density and chick production thus included the following explanatory fixed factors: the proportion of old forest in interaction with the proportion of mid to high productive forest, SiD in interaction with mean forest stand size, and species in three-way interaction with the two interactions above. Based on evaluation of the residuals from the global models, we transformed estimates of adult density ($\ln(\text{Adult density} + 1)$), but kept chick production untransformed in the models.

We used AIC, corrected for small sample size (AICc, Burnham and Anderson 2002) to rank alternative candidate models. A difference in AICc ($\Delta AICc$) of less than two between alternative models suggests that these should be considered as equally credible (Burnham and Anderson 2002). From the AICc-values we also calculated AICc-weights (AICc-w), which is the probability that a model is the best, given the set of candidate models (Burnham and Anderson 2002). The AICc was calculated from models fitted using maximum likelihood (ML), whereas parameter estimates are based on models fitted with restricted maximum likelihood (REML) (Zuur et al. 2009). Effects of explanatory variables were estimated with posterior resampling (MCMC, $n = 10,000$) of the parameter estimates from the fitted model, where the 2.5 and 97.5 % quantiles indicated the 95 % confidence interval (Bates et al. 2011b). We also quantified the proportion of variance between study areas that were captured by the fixed effects following Singer (1998). This procedure compares the reduction of the variance in a random factor (i.e. study area) between two models with different fixed effect structure. We compared the variance attributed to study area as random factor from a model with grouse species only, to the more complex model that were chosen as best based on the AICc-values. All statistical analyses were

done in R version 2.13.1 (R Development Core Team 2011).

Multicollinearity among explanatory variables, which were present among some of our landscape variables (Appendix 2 in in Supplementary Material) can cause biased parameter estimates (e.g. Graham 2003; Dormann et al. 2013). However, linear models are in general quite robust with respect on correlation among the explanatory variables (Freckleton 2011). Moreover, evaluating variables importance using AICc is thought to find the combination of explanatory variables that best capture the variation in the response variable by comparing different combinations of the explanatory variables, thereby better addressing multicollinearity than stepwise procedures (e.g. Burnham and Anderson 2002; Johnson and Omland 2004). In addition, because we were interested in how multiple characteristics that described the landscape predicted demographic traits of forest grouse, we choose to keep all explanatory variables. Nevertheless, we assessed the influence of multicollinearity among our explanatory variables by comparing parameter estimates from several models. For both adult density and chick per adult female, the parameter estimates were stable across candidate models with $\Delta\text{AICc} < 2$. Moreover, leaving out one of the correlated variable from a model with multicollinearity did not affect the direction of the relationship with the demographic trait, which supports that the parameter estimates from our models about spatial variation in demographic traits not suffers from bias due to multicollinearity. Following Freckleton (2011), we chose to report estimates from the chosen model instead of for instance model averaging.

Our data on spatial variation in grouse demographic rates originated from time series within each study area. Such data often have temporal and/or spatial autocorrelation among the observations, which can affect parameter estimates. To assess whether temporal autocorrelation was present in our models, we tested the 1 year lag autocorrelation of the residuals of the highest ranked model, but did not find evidence of significant autocorrelation within a 95 % confidence interval for capercaillie adult density and chick production. For black grouse we found 3 incidents out of 13 (the total number of time series) for adult density and 2 incidents out of 13 for chick production within the confidence interval. Regarding spatial autocorrelation, we first calculated area-specific mean

residual values for the highest ranked model on adult density and chick production separately. We then regressed the pairwise differences in residual value for all areas against the corresponding distance between the center point of the areas. Any positive trend in difference in residual value with distance would suggest that areas close to each other were more similar than areas further away. For neither adult survival ($r_p = 0.011$, $t = 0.09$, $df = 76$, $P = 0.926$) or chick production ($r_p = -0.168$, $t = -1.49$, $df = 76$, $P = 0.141$), we found evidence for significant trends in residual difference with distance between areas. Combined with the lack of evidence of temporal autocorrelation, this indicates that our models were not subject to biases in the parameter estimates due to autocorrelation among the observations.

Results

The average adult density estimates within the areas ranged from 1.2 to 4.7 capercaillie per km², with a mean of 2.6 per km². Adult density of black grouse ranged from 1.5 to 4.8 per km² with mean 3.1 per km². Chick production of capercaillie and black grouse ranged from 0.3 to 2.8 (mean 1.8) and 0.9 to 4.3 (mean 2.6) chick per female, respectively (Appendix 1 in Supplementary Material). Areas with no registered chick production for one of the species in a given year had less than 11 adult observations that year.

The highest ranked model explaining variation in adult density of capercaillie and black grouse between the study areas included proportion of old forest, the proportion of mid to high productive forest, as well as the interaction between old forest and mid to high productive forest and the main effect of species (Table 3). Several alternative models had ΔAICc less than two, and only one was quite close to the highest ranked model (Table 3). The second best model ($\Delta\text{AICc} = 0.18$) also included the interaction between grouse species and old forest, but this interaction was not included in any of the other candidate models with $\Delta\text{AICc} < 2$. We therefore did not consider this model to be important for explaining the variation in adult density among the 13 areas, and we retained the highest ranked model as the most likely among the candidate models. The highest ranked model (Table 3) suggested a complex relationship between old forest,

Table 3 The highest ranked models according to AICc-values explaining the variation in adult density and chick production of capercaillie and black grouse

Model specification	ΔAICc	AICc-w
Adult density		
Species + mid to high productive forest $\times$ old forest	0.00	0.067
Species $\times$ old forest + mid to high productive forest $\times$ old forest	0.18	0.061
Mid to high productive forest $\times$ old forest	1.19	0.037
Species $\times$ mid to high productive forest + mid to high productive forest $\times$ old forest	1.51	0.032
Species + mid to high productive forest + SiD	1.51	0.032
Species + mid to high productive forest	1.85	0.027
Species + mid to high productive forest $\times$ old forest + mean stand size	1.96	0.025
Chick production		
Species $\times$ SiD + old forest	0.00	0.084
Species + SiD + old forest	0.02	0.083
Species $\times$ old forest + SiD	1.91	0.032
Species $\times$ SiD + species $\times$ old forest	1.96	0.031

For variables description, see Table 1. An $\times$ indicates interaction between two variables, and includes also the main effect of the two interacting variables. Only candidate models with $\Delta\text{AICc} < 2$ is presented. For specification about the full global model, see “Methods” section

mid to high productive forest and the adult density of the two species. The positive estimate of the interaction between mid to high productive forest and old forest ($\beta = 9.19 \pm 3.90$ SE) on adult density suggested that the effect of old forest was influenced by the proportion of mid to high productive forest in an area. With low availability of mid to high productive forest, there were no evident relationships between proportion of old forest and adult density (Fig. 2a). For areas with a large proportion of mid to high productive forest, there were a strong positive relationship between old forest and adult density (Fig. 2b). This suggests that access to old forest is important if a large proportion of the forest is of mid to high-productivity (Fig. 2a, b). Similarly, access to mid to high-productive areas is more beneficial if a large proportion of the forest is in old age-classes.

Among the candidate models to explain variation in chick production of forest grouse, four models had

$\Delta\text{AICc} < 2$ (Table 3). Two explanatory variables appeared in all these four models; habitat diversity (SiD) and proportion of old forest. The highest ranked model included SiD, proportion of old forest, species, and the interaction between SiD and species (Table 3). The second highest ranked model ($\Delta\text{AICc} = 0.02$) did not include the interaction between SiD and species (Table 3). Based on the appearance of the variables and interaction in several of the candidate models, we chose the highest ranked model to be most likely to explain variation in chick production among forest grouse. According to this model the chick production was negatively related to the proportion of old forest, similarly for both species ($\beta = -5.73 \pm 2.30$ SE, Fig. 2c). Moreover, the chick production was positively related to SiD for both species, but this relationship was weaker for capercaillie compared to black grouse (capercaillie: $\beta = 5.68 \pm 2.52$ SE, black grouse: $\beta = 8.57 \pm 2.17$, Fig. 2d).

Year and study area as random factors explained 35.2 and 13.2 %, respectively, of the variance in chick production in the highest ranked models, whereas for adult density, year (4.9 %) and study area (4.9 %) had a much less explanatory effect. This suggests a much larger temporal and spatial variation in chick production than in adult density. The highest ranked model captured 81.0 % of the variation in adult density among study areas compared to a model with only grouse species as a fixed factor (both models had area identity and year as random factor). For chick production, the corresponding value was 41.6 %. This suggests that our model captured large proportions of the spatial variation in grouse demographic rates, but that our variables describing forest characteristics better captured spatial variation in adult density compared to chick production.

Discussion

Based on the first large scale estimates of time series of population density of forest grouse (black grouse and capercaillie) in Norway, we have related adult density and chick production to habitat variables at the landscape scale. The main effect of species was included in all except one of the highest ranked models (Table 3), which documents differences in adult density and chick production between the two species (black grouse being the smaller, more abundant and

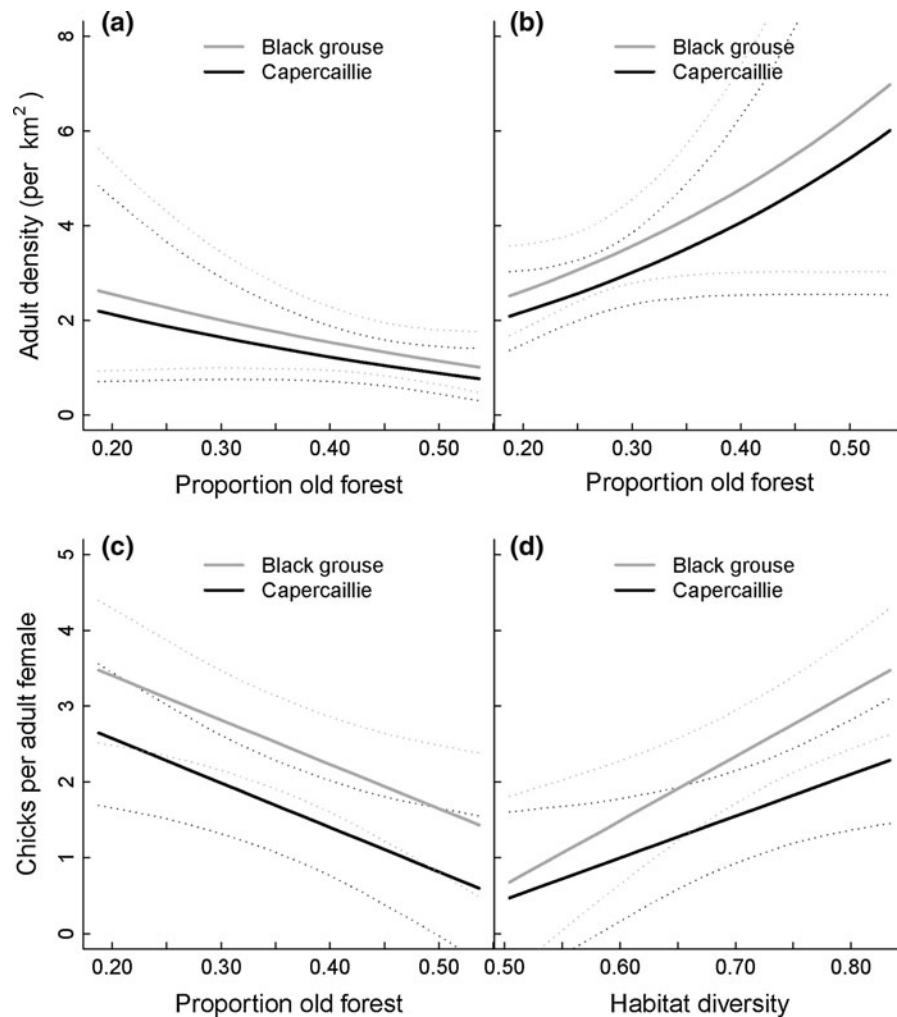


Fig. 2 Predicted relationship between landscape characteristics and demographic traits in capercaillie (*black lines*) and black grouse (*gray lines*), based on the in Table 3. **a** The relationship between adult density and the proportion old forest recognized by a low proportion of mid to high productive forest (i.e. the 10 % quantile of mid to high productive forest among the study areas). **b** The relationship between adult density and the proportion old forest with a high proportion of mid to high

productive forest (i.e. the 90 % quantile of mid to high productive forest, **c** The relationship between chick production and the proportion of old forest. **d** The relationship between chick production and habitat diversity (SiD). The *thin lines* represent 95 % confidence interval based on the 2.5 and 97.5 % quantiles from 10,000 MCMC resamplings from the posterior distribution of the parameter estimates

fecund species). In our hypotheses regarding relationships between demographic traits and landscape characteristics, we found support for a positive relationship between adult density and proportion of old forest (but depending on forest productivity, Fig. 2a, b) and a positive relationship between landscape heterogeneity and chick production (Fig. 2d). However, we did not find evidence for a positive relationship between proportion of old forest and chick production (Fig. 2c), nor that habitat

heterogeneity affected adult density. The few significant interactions between species and landscape variables suggest that the influence of landscape characteristics on demographic traits is similar for black grouse and capercaillie. However, as hypothesized, adult density and chick production were differently related to the forest characteristic variables. This suggests that forest grouse depend on different forest types during their life cycles (Sirkiä et al. 2010; Bollmann et al. 2011), and contributes to the

increasing evidence that habitat diversity is an important factor for variation in demographic traits, life history and spatial variation in population abundance (e.g. Wang et al. 2006; Oliver et al. 2010; Herfindal et al. 2013; G  thlin et al. 2013).

Winter is often the critical period for adult survival of birds and mammals living in temperate ecosystems (Newton 1998). Animals often seek habitat that provides a combination of access to food, cover from predation, and shelter from harsh weather. The importance of a combination of high productive forest and old forest for adult density of both capercaillie and black grouse (Table 3; Fig. 2a, b) is in line with this. Natural causes of adult mortality (i.e. not hunting) of both species is highest during winter, and mainly related to access to forage, weather conditions and predation (Wegge et al. 1987; Kangas and Kurki 2000; Kurki et al. 2000). Mid to high productive conifer forest is characterized by rich moist soil and rich cover at these latitudes, and meets the demands for both capercaillie and black grouse throughout most of the year (Miettinen et al. 2005, 2010; Wegge and Rolstad 2011). However, the positive interaction between mid to high productive forest and proportion of old forest suggests that access to old forest is important when a large proportion of the forest is in mid to high productive areas. Findings from the national forest inventory suggest that old forest may have as much as 25 % bilberry (*Vaccinium myrtillus*) cover (National Forest Inventory 2008), and considerably more in high compared to low productive forests (Bj  rneraas et al. 2012). Bilberry is an important food resource, particularly during the snow-free period of the year (Wegge and Kastdalen 2008; Ludwig et al. 2010). Old, high-productive forest, in particular mixed conifer forests with Scots pine (capercaillie), bilberry and birch (black grouse), provides a combination of forage during winter (Willebrand 1988; Gjerde and Wegge 1989) as well as cover and shelter from predators and harsh weather (Spids   et al. 1997; Thiel et al. 2007).

Recruitment (i.e. clutch size and juvenile survival) is often one of the most important demographic trait for population growth of birds with short life histories (S  ther and Bakke 2000), and this is also found for our study species (Wegge and Kastdalen 2007; Ludwig et al. 2010). It is therefore important to detect essential habitats during this life stage. Leks and nests seem to be placed more randomly within the forest matrix for both capercaillie and black grouse, and are often found

in thinned stands, younger plantations as well as mixed conifer with deciduous trees and bogs (Rolstad and Wegge 1987; Miettinen et al. 2005; Rolstad et al. 2007; Wegge and Rolstad 2011). In concordance with this, we found that habitat heterogeneity was important for chick production in both species (Table 3; Fig. 2d), but more so for black grouse than capercaillie. Consequently, a heterogeneous landscape consisting of young and old both Norway spruce and Scots pine forest, mixed forests, deciduous forest and bog, appears to be important for recruitment. High habitat diversity also involves edges between habitat types. Edges can be favorable for forest grouse (Kurki et al. 2000), possibly because they offer shelter, escape routes, a diverse nutrition access within short distances, and even turbulent winds making it harder for predators (e.g. *Vulpes vulpes*) to smell their prey (Conover 2007).

Chick production of capercaillie and black grouse was negatively related to the proportion of forest older than 80 years (Fig. 2c). Such forest is in general rich on bilberry, which is found to be important for forest grouse (Storch 1995; Sel  s 2001; Summers et al. 2004a; Gregersen and Gregersen 2008) and particularly for chick survival (Ludwig et al. 2010). However, our results, together with recent findings (Miettinen et al. 2008; Wegge and Rolstad 2011) suggest that old forest is not associated with large broods during late summer. In late summer, the number of chicks per adult hen is a factor of the number of eggs laid, nest predation, and chick survival after hatching, with nest and chick predation being the most important (Wegge and Storaas 1990; Wegge and Kastdalen 2007). The role of forest characteristics on predator abundance and predation rates, studied at local scale by Summers et al. (2004b) and Wegge and Rolstad (2011), should be further investigated at large spatial scale to fully understand how the landscape affects forest grouse recruitment.

All of our study areas were large enough to meet the area requirements of at least one capercaillie annual home range (1–12 km², Rolstad et al. 1988; Storch 1995). Even though the density estimates are surveyed during late summer they are a result of the ability of the landscape to produce and support viable populations of capercaillie and black grouse throughout the year. Our results support that habitat requirements of black grouse and capercaillie are quite diverse when including several life history stages, and that

previously suggested habitat requirements based on for instance only chick production were too conservative. We conclude that both species persist in relative high densities in Norwegian forests despite a rapidly and constantly changing landscape through forestry.

Our results, together with recent findings (Lande et al. 2010; Sirkiä et al. 2011; Wegge and Rolstad 2011) suggest that forest grouse have less specific habitat requirements than previously suggested (Seiskari 1962; Swenson and Angelstam 1993; Wallgren et al. 2009). Large homogenous forests in late successional stages were quite common in the 1950–1960s, and as a consequence capercaillie utilized such forests. As of today, the boreal forest in Scandinavia is rather heterogeneous, but still holds viable populations of both capercaillie and black grouse (Kålås et al. 2010), nevertheless showing a slowly decreasing population trend (Gegersen and Gegersen 2009). Population dynamics of capercaillie and black grouse are described to fluctuate synchronously over large spatial scales (Ranta et al. 1995; Kvasnes et al. 2010), and factors like predation, weather conditions (particularly temperature, Ludwig et al. 2010) and nutrition availability during chick stage affect both species to a similar extent. It seems thus quite reasonable to consider an active management for both species rather than single species management regimes, and our results indicate that old mid to high-productive forests are particularly important for the adult segment. Moreover, it seems that our large-scale variables describing forest landscapes were more suitable to capture adult densities than chick production. Consequently, more research is needed to explore the factors affecting recruitment to aid management of forest grouse. This can be factors that operate at a finer spatial scale, making large-scale management less appropriate for chick production compared to adult density.

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